Linear Algebra — Lecture 1

Systems of Linear Equations, Row Reduction,
Vector Equations and Matrix Equations

Friedberg–Insel–Spence, *Linear Algebra* 4e, §1.4, §3.3–§3.4
D. C. Lay, *Linear Algebra and Its Applications* 4e, §1.1–§1.4

Topics. Systems of linear equations, row reduction and echelon forms, vector equations and matrix equations.

How the two texts divide the work. FIS (the main text) develops the theory of an **abstract vector space over an arbitrary field** $\mathbb{F}$; Lay supplies concrete examples in $\mathbb{R}^n$ and the computational intuition. Mapping for this lecture:

Topic	Lay 4e (printed)	Friedberg 4e (printed)
Systems of linear equations	§1.1 (p. 2)	§1.4 (pp. 24–28), §3.3 (p. 168)
Row reduction, echelon form	§1.2 (p. 12)	§3.4 (pp. 182–198)
Vector equations, linear combinations	§1.3 (p. 24)	§1.4 (pp. 24–31)
Matrix equation $A\boldsymbol{x} = \boldsymbol{b}$	§1.4 (p. 34)	§3.3 (pp. 168–181)

Two questions that drive the whole lecture: (1) *Existence* — is the system consistent? (2) *Uniqueness* — if consistent, is the solution unique?

How to read: use Lay first to get the algorithm under your fingers, then Friedberg to see why each step is legitimate.

All page numbers here are printed pages. For Lay, PDF page = printed page +17; for Friedberg, PDF page = printed page +15.

1 Systems of Linear Equations

Definition 1.1 (Linear equation and linear system)**.** A **linear equation** in the variables $x_1, \dots, x_n$ is one that can be written as

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b,$$

where $a_1, \dots, a_n, b$ are real (or complex) constants. A collection of such equations is a **system of linear equations** (or a *linear system*).

A list $(s_1, \dots, s_n)$ is a **solution** if it satisfies *every* equation simultaneously; the set of all solutions is the **solution set**. Two systems are **equivalent** if they have the same solution set. A system is **consistent** if it has at least one solution, and **inconsistent** otherwise.

Proposition 1.1 (Only three possibilities)**.** *A system of linear equations has either*

(i) no solution,
(ii) exactly one solution, or
(iii) infinitely many solutions.

Definition 1.2 (Matrices of a system)**.** For the system

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = b_1 \\ \quad\vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n = b_m \end{cases}$$

the **coefficient matrix** and the **augmented matrix** are

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}, \qquad \left[\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & b_1 \\ \vdots & \ddots & \vdots & \vdots \\ a_{m1} & \cdots & a_{mn} & b_m \end{array}\right].$$

An $m \times n$ matrix has m rows and n columns; the **size** is always *rows first.*

Definition 1.3 (Elementary row operations)**.**

1. **Replacement:** add to one row a multiple of another row;
2. **Interchange:** swap two rows;
3. **Scaling:** multiply all entries of a row by a nonzero constant.

Remark 1.1. Each of the three operations is *reversible* and therefore produces an **equivalent** system — the solution set is unchanged. This is the entire justification for Gaussian elimination.

Remark 1.2 (Why row operations are legitimate — Friedberg's proof)**.** Lay merely asserts that the three operations preserve the solution set; Friedberg turns this into a theorem (Chapter 3). Each elementary row operation amounts to **left multiplication by an elementary matrix** (§3.1, p. 148), which gives

Theorem 3.13 (p. 182). Let $A\boldsymbol{x} = \boldsymbol{b}$ be a system of m equations in n unknowns and let C be an invertible $m \times m$ matrix. Then the system $(CA)\boldsymbol{x} = C\boldsymbol{b}$ is **equivalent** to $A\boldsymbol{x} = \boldsymbol{b}$ (same solution set).

This is the rigorous answer to "why may I freely perform row operations?".

2 Row Reduction and Echelon Forms

Definition 2.1 (Echelon form)**.** A matrix is in **echelon form** (row echelon form) if

1. all nonzero rows are above any rows of all zeros;
2. each leading entry of a row is in a column to the right of the leading entry of the row above it;
3. all entries in a column below a leading entry are zero.

It is in **reduced echelon form** if, in addition,

4. the leading entry in each nonzero row is 1;
5. each leading 1 is the only nonzero entry in its column.

Theorem 2.1 (Uniqueness of the reduced echelon form)**.** *Every matrix is row equivalent to one and only one reduced echelon matrix.*

Remark 2.1. Echelon forms are *not* unique, but the *reduced* echelon form is. This is what makes later notions (rank, pivot columns) well defined.

Definition 2.2 (Pivot)**.** A **pivot position** in A is a location in A that corresponds to a leading 1 in the reduced echelon form of A. A **pivot column** is a column of A containing a pivot position. Variables corresponding to pivot columns are **basic variables**; the remaining ones are **free variables**.

Definition 2.3 (Row reduction algorithm)**. Forward phase** (produces an echelon form):

1. Begin with the leftmost nonzero column; it is a pivot column, with the pivot at the top.
2. Use row replacement to create zeros below the pivot.
3. Cover the row containing the pivot and all rows above it; repeat steps 1–2 on the remaining submatrix until no nonzero rows remain.

Backward phase (produces the reduced echelon form):

4. Starting with the rightmost pivot, use row replacement to create zeros *above* it.
5. Scale each pivot row so that its pivot becomes 1.

Theorem 2.2 (Existence and Uniqueness Theorem)**.** *A linear system is consistent if and only if the rightmost column of the augmented matrix is* ***not*** *a pivot column — equivalently, if and only if an echelon form of the augmented matrix has no row of the form*

$$\left[\begin{array}{ccc|c} 0 & \cdots & 0 & b \end{array}\right] \qquad (b \neq 0).$$

If a system is consistent, then its solution set contains either

(i) a unique solution, when there are no free variables, or
(ii) infinitely many solutions, when there is at least one free variable.

Example 2.1 (A complete row reduction)**.** Solve

$$\begin{cases} x_1 - 2x_2 + x_3 = 0 \\ 2x_2 - 8x_3 = 8 \\ -4x_1 + 5x_2 + 9x_3 = -9 \end{cases}$$

Approach. Do not set out to "solve" the system: reduce the augmented matrix to reduced echelon form instead, handling one pivot column at a time and doing only two things per column — scale the pivot to 1, then clear both above and below it.

$$\left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ -4 & 5 & 9 & -9 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & -3 & 13 & -9 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & -7 & 8 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & 3 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 29 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{array}\right].$$

Every column is a pivot column, so there are no free variables. The solution is therefore unique: $(x_1, x_2, x_3) = (29, 16, 3)$.

Remark 2.2 (Parametric description)**.** When free variables occur, solve each pivot-variable equation for its basic variable in terms of the free ones. Writing $x_3 = t$ etc. gives the general solution in parametric (vector) form; this is the standard way to describe an infinite solution set.

3 Vector Equations

Vectors in $\mathbb{R}^n$ are $n \times 1$ column matrices. For $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$ and $c \in \mathbb{R}$, addition and scalar multiplication are defined componentwise. Geometrically, $\boldsymbol{u} + \boldsymbol{v}$ is given by the **parallelogram rule**, and $c\boldsymbol{v}$ is a scaling (reversing direction when $c < 0$).

Definition 3.1 (Linear combination)**.** Given $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p \in \mathbb{R}^n$ and scalars $c_1, \ldots, c_p$, the vector

$$\boldsymbol{y} = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + \cdots + c_p\boldsymbol{v}_p$$

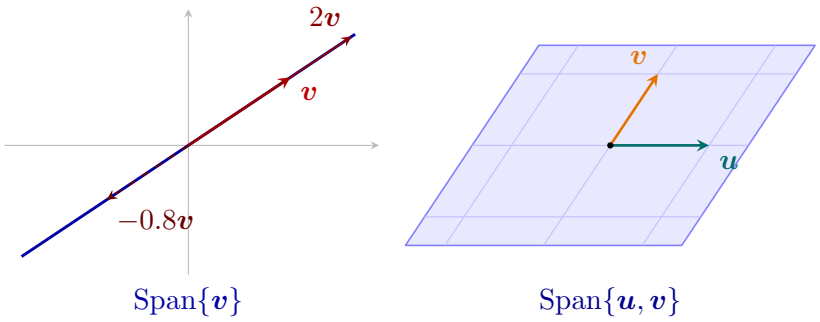
is a **linear combination** of $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p$ with **weights** $c_1, \ldots, c_p$.

Definition 3.2 (Span)**.** If $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p \in \mathbb{R}^n$, the set of all linear combinations of them is denoted $\mathrm{Span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p\}$ and is called the subset of $\mathbb{R}^n$ **spanned** (or generated) by $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p$.

Proposition 3.1 (The key translation)**.** *For $\boldsymbol{b} \in \mathbb{R}^n$, the following are equivalent:*

(i) $\boldsymbol{b} \in \mathrm{Span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p\}$*;*
(ii) the vector equation $x_1\boldsymbol{v}_1 + \cdots + x_p\boldsymbol{v}_p = \boldsymbol{b}$ *has a solution;*
(iii) the linear system with augmented matrix $\left[\begin{array}{ccc|c} \boldsymbol{v}_1 & \cdots & \boldsymbol{v}_p & \boldsymbol{b} \end{array}\right]$ *is consistent.*

Remark 3.1 (Geometry)**.** $\mathrm{Span}\{\boldsymbol{v}\}$ ($\boldsymbol{v} \neq \boldsymbol{0}$) is a line through the origin; $\mathrm{Span}\{\boldsymbol{u}, \boldsymbol{v}\}$ with $\boldsymbol{u}, \boldsymbol{v}$ non-proportional is a plane through the origin. Every span contains $\boldsymbol{0}$ (take all weights zero).



(a) $\mathrm{Span}\{\boldsymbol{v}\}$ is a line through the origin; (b) when $\boldsymbol{u}, \boldsymbol{v}$ are not proportional, $\mathrm{Span}\{\boldsymbol{u}, \boldsymbol{v}\}$ is a plane through the origin.

Remark 3.2 (The abstract version: Friedberg §1.4, pp. 24–31)**.** Lay works inside $\mathbb{R}^n$; Friedberg defines the same notions for a **vector space** V **over an arbitrary field** $\mathbb{F}$:

- For nonempty $S \subseteq V$, a vector v is a **linear combination** of vectors of S if $v = a_1u_1 + \cdots + a_nu_n$ for some $u_1, \ldots, u_n \in S$ and $a_1, \ldots, a_n \in \mathbb{F}$; the a_i are the **coefficients**.
- The **span** $\mathrm{Span}(S)$ is the set of all linear combinations of vectors in S, with the convention $\mathrm{Span}(\emptyset) = \{\boldsymbol{0}\}$.
- **Theorem 1.5** (p. 30): $\mathrm{Span}(S)$ is a **subspace** of V, and every subspace containing S also contains $\mathrm{Span}(S)$ — i.e. $\mathrm{Span}(S)$ is the **smallest subspace containing** S.
- If $\mathrm{Span}(S) = V$, then S **generates** (or spans) V.

Worth comparing with the book's examples (pp. 31–32): $(1,1,0), (1,0,1), (0,1,1)$ generate $\mathbb{R}^3$; three polynomials generate $P_2(\mathbb{R})$; four matrices generate $M_{2\times 2}(\mathbb{R})$, whereas three matrices all having equal diagonal entries do *not* generate $M_{2\times 2}(\mathbb{R})$.

4 The Matrix Equation $A\boldsymbol{x} = \boldsymbol{b}$

Definition 4.1 (The product $A\boldsymbol{x}$)**.** If A is an $m \times n$ matrix with columns $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$, and if $\boldsymbol{x} \in \mathbb{R}^n$, then

$$A\boldsymbol{x} = x_1\boldsymbol{a}_1 + x_2\boldsymbol{a}_2 + \cdots + x_n\boldsymbol{a}_n.$$

That is, $A\boldsymbol{x}$ *is the linear combination of the columns of A with weights given by the entries of $\boldsymbol{x}$.*

Remark 4.1. Read every occurrence of $A\boldsymbol{x}$ as "a linear combination of the columns of A." This single viewpoint is the one that survives into abstract vector spaces later in the course.

Theorem 4.1 (Three equivalent formulations)**.** *Let A be $m \times n$ with columns $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$ and let $\boldsymbol{b} \in \mathbb{R}^m$. Then the matrix equation $A\boldsymbol{x} = \boldsymbol{b}$, the vector equation $x_1\boldsymbol{a}_1 + \cdots + x_n\boldsymbol{a}_n = \boldsymbol{b}$, and the linear system with augmented matrix* $\left[\begin{array}{ccc|c} \boldsymbol{a}_1 & \cdots & \boldsymbol{a}_n & \boldsymbol{b} \end{array}\right]$ *all have the **same solution set**.*

Theorem 4.2 (Four equivalent statements)**.** *Let A be an $m \times n$ matrix. The following are logically equivalent: for a given A, either all are true or all are false.*

(a) For each $\boldsymbol{b} \in \mathbb{R}^m$, the equation $A\boldsymbol{x} = \boldsymbol{b}$ has a solution.
(b) Each $\boldsymbol{b} \in \mathbb{R}^m$ is a linear combination of the columns of A.
(c) The columns of A span $\mathbb{R}^m$.
*(d) A has a pivot position in **every row**.*

Theorem 4.3 (Consistency and structure of the solution set)**.** *Let A be $m \times n$ and $\boldsymbol{b} \in \mathbb{R}^m$. Then:*

(i) $A\boldsymbol{x} = \boldsymbol{b}$ is consistent iff $\boldsymbol{b}$ is a linear combination of the columns of A;
(ii) if $A\boldsymbol{x} = \boldsymbol{b}$ is consistent, its solution set is the translate of the solution set of $A\boldsymbol{x} = \boldsymbol{0}$:

$$\{\boldsymbol{x}\} = \boldsymbol{p} + \{\boldsymbol{x}_h : A\boldsymbol{x}_h = \boldsymbol{0}\},$$

where $\boldsymbol{p}$ is any particular solution.

Definition 4.2 (Identity matrix)**.** The **identity matrix** I_n is the $n \times n$ matrix with 1's on the diagonal and 0's elsewhere; it satisfies $I_n\boldsymbol{x} = \boldsymbol{x}$ for every $\boldsymbol{x} \in \mathbb{R}^n$.

Remark 4.2 (Friedberg's rigorous version: §3.3, pp. 168–181)**.** Friedberg states the corresponding facts in terms of **rank**, over an arbitrary field $\mathbb{F}$:

- **Theorem 3.11** (p. 174): $A\boldsymbol{x} = \boldsymbol{b}$ is consistent iff $\operatorname{rank}(A) = \operatorname{rank}(A|\boldsymbol{b})$.
- **Theorem 3.8** (p. 171): for the homogeneous system $A\boldsymbol{x} = \boldsymbol{0}$, the solution set is $K = N(L_A)$, a **subspace** of $\mathbb{F}^n$ of dimension $n - \operatorname{rank}(A)$.
- **Theorem 3.9** (p. 172): if K is the solution set of $A\boldsymbol{x} = \boldsymbol{b}$ and K_H that of $A\boldsymbol{x} = \boldsymbol{0}$, then for any particular solution $\boldsymbol{s}$, $K = \{\boldsymbol{s}\} + K_H$.
- **Theorem 3.10** (p. 174): for n equations in n unknowns, A is invertible iff there is exactly one solution, namely $\boldsymbol{x} = A^{-1}\boldsymbol{b}$.

Theorem 4.3 (Friedberg's Theorem 3.9) and Lay's Theorem 6 (§1.5, p. 46) are the same statement, but Friedberg's phrasing (particular solution + homogeneous solution space) is the standard language used later on. These results presuppose "rank", which Friedberg only defines in §3.2 — **master Lay's pivot criterion first** and come back to compare once the course reaches rank.

5 A Worked Example

Example 5.1 (Tying the four viewpoints together: consistency, pivots, spanning; This example was written for this lecture; the methods are Lay §1.1–§1.4)**.** Let

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & -1 \\ 1 & 4 & 1 \end{bmatrix}, \qquad \boldsymbol{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix},$$

and write the three columns of A as $\boldsymbol{v}_1 = (1, 0, 1)^\top$, $\boldsymbol{v}_2 = (3, 1, 4)^\top$, $\boldsymbol{v}_3 = (2, -1, 1)^\top$. Find the condition for $A\boldsymbol{x} = \boldsymbol{b}$ to be consistent, decide whether the solution is then unique, and decide whether the three columns span $\mathbb{R}^3$.

Approach. When the question is "for which $\boldsymbol{b}$ is the system consistent", do not first guess a form for $\boldsymbol{b}$ — reduce the coefficient matrix to reduced echelon form instead. That one echelon form answers all three questions at once: whether the augmented column creates a contradictory row (consistency), how many columns are not pivots (degrees of freedom), and which dimension the columns' span is missing.

(1) Find the condition for $A\boldsymbol{x} = \boldsymbol{b}$ to be consistent. Row reduce the augmented matrix to

reduced echelon form (Lay §1.2, p. 12):

$$\left[\begin{array}{ccc|c} 1 & 3 & 2 & b_1 \\ 0 & 1 & -1 & b_2 \\ 1 & 4 & 1 & b_3 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 3 & 2 & b_1 \\ 0 & 1 & -1 & b_2 \\ 0 & 1 & -1 & b_3 - b_1 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 5 & b_1 - 3b_2 \\ 0 & 1 & -1 & b_2 \\ 0 & 0 & 0 & b_3 - b_1 - b_2 \end{array}\right].$$

The last row reads $0 = b_3 - b_1 - b_2$, so the system is **consistent exactly when** $b_3 = b_1 + b_2$; the case $b_3 \neq b_1 + b_2$ is the "no solution" branch.

(2) Is the solution unique when the system is consistent? The pivot columns of the reduced form are columns 1 and 2, while the number of unknowns is $n = 3$, so x_3 is free:

$$x_1 = b_1 - 3b_2 - 5x_3, \qquad x_2 = b_2 + x_3 \qquad (x_3 \text{ arbitrary}),$$

hence there are **infinitely many solutions**. That is the trichotomy of Lay §1.1, p. 2 in action.

(3) Do the three columns span $\mathbb{R}^3$? Column 3 is not a pivot column, so $\boldsymbol{v}_3$ is a linear combination of the first two; the reduced form reads it off:

$$\boldsymbol{v}_3 = 5\boldsymbol{v}_1 - \boldsymbol{v}_2 \qquad \big(\text{check: } 5(1,0,1)^\top - (3,1,4)^\top = (2,-1,1)^\top\big),$$

so

$$\operatorname{Span}\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\} = \operatorname{Span}\{\boldsymbol{v}_1, \boldsymbol{v}_2\} = \{(b_1, b_2, b_3)^\top : b_3 = b_1 + b_2\},$$

a **plane** through the origin in $\mathbb{R}^3$; the three vectors therefore do **not** span $\mathbb{R}^3$.

In one sentence: the same echelon form answers three questions – whether the augmented column creates a contradictory row decides **consistency**, the number of pivot columns decides the **degrees of freedom**, and which columns are *not* pivots decides **which dimension the span is missing**. Friedberg writes the first of these as an equation between ranks (Theorem 3.11, §3.3, p. 174).

Remark 5.1 (Not covered here – these belong to later lectures)**.**

- **Fields, vector spaces, subspaces**: Friedberg §1.2–§1.3, p. 6, p. 16; Lay §4.1, p. 190. Belongs to **Lecture 2**.
- **Linear independence, bases, dimension**: Friedberg §1.5–§1.6, p. 35, p. 42; Lay §4.3–§4.5, p. 208, p. 216, p. 225. Also **Lecture 2**.
- **Rank**: Friedberg §3.2; Lay §4.6, p. 230. This lecture only uses the equivalent phrasing "number of pivots"; come back to rank when the course reaches it.
- **Dual spaces, bilinear forms, the Cayley–Hamilton theorem**: absent from Lay; Friedberg is the only source.

6 Checklist and Exercises

By the end of this lecture you should be able to:

1. Write the augmented matrix of a system and reduce it to reduced echelon form without errors.
2. Read off from an echelon form whether the system has no solution, one solution, or infinitely many, and produce the general (parametric) solution.
3. Explain the correspondence between pivot columns, basic variables and free variables.
4. Translate "is $\boldsymbol{b} \in \operatorname{Span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p\}$?" into a question about consistency of an augmented matrix.

5. State the definition of $A\boldsymbol{x}$ and explain Theorem 4.1.
6. Use condition (d) of Theorem 4.2 to decide whether the columns of A span $\mathbb{R}^m$.

Exercises (Lay 4e). §1.1 p. 10 · §1.2 p. 21 · §1.3 p. 32 · §1.4 p. 40. Each section also ends with *Practice Problems* before the exercise set — do those first.

How to use the two texts together:

- **Lay for computation, Friedberg for proof.** The same result carries different numbers in the two books; these notes give both.
- Friedberg works over an **arbitrary field** $\mathbb{F}$, not just $\mathbb{R}$. When a problem involves $\mathbb{C}$ or $\mathbb{Z}_p$, only Friedberg's statements apply — Lay's conclusions do not transfer automatically.
- The core chain of this lecture: *augmented matrix* $\to$ *echelon form* $\to$ *pivots decide existence and uniqueness* $\to$ *pivot and free columns describe* Span.